CHAPTER 13

MINERAL AND ENERGY RESOURCES

CHAPTER ESSENCE AT A GLANCE:
Master India metallic and non-metallic mineral endowments, the three principal mineral belts, conventional fossil fuels (coal, petroleum, natural gas), nuclear energy, and renewable non-conventional energy installations.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Mineral Classification and Belts:** Metallic minerals are ferrous (iron ore, manganese) or non-ferrous (bauxite, copper). Non-metallic minerals include mica and limestone. Three major belts: North-Eastern Plateau (Chotanagpur, mineral heartland), South-Western Plateau (Karnataka-Goa, ferrous metals), and North-Western Region (Aravallis, non-ferrous metals).

• **Iron Ore and Manganese Wealth:** India possesses haematite and magnetite iron ore deposits in Odisha (Mayurbhanj, Keonjhar), Jharkhand (Singhbhum), and Chhattisgarh (Bailadila). Odisha is the leading manganese producer.

• **Conventional Energy Resources:** Coal is divided into Gondwana coal (200 million years old, Damodar valley, 98% of reserves) and Tertiary coal (young, lignite in Neyveli). Petroleum originates in Mumbai High offshore, Digboi (oldest in Assam), and Ankleshwar in Gujarat.

• **Nuclear and Renewable Energy:** Nuclear power utilizes Uranium (Jaduguda) and Thorium (Monazite beach sands of Kerala) at Tarapur, Rawatbhata, Kalpakkam, and Narora. Renewable energy includes Kamuthi solar park and Muppandal wind cluster in Tamil Nadu.

ESSENTIAL GEOGRAPHICAL MODELS & LANDFORMS

$$\text{North-Eastern Plateau Mineral Belt} \rightarrow \text{Chotanagpur} \iff \{\text{Iron Ore, Coal, Manganese, Bauxite, Mica}\}$$
Chotanagpur Mineral Heartland Complex

$$\text{South-Western Mineral Belt} \rightarrow \text{Karnataka and Goa} \iff \{\text{Ferrous Iron Ore, Limestone, Monazite Sands}\}$$
Peninsular South-Western Mineral Domain

$$\text{Conventional Fossil Energy} \iff \text{Gondwana and Tertiary Coal} + \text{Bombay High and Assam Petroleum}$$
Conventional Fossil Fuel Energy Mix

$$\text{Non-Conventional Renewable Energy} \rightarrow \text{Tarapur Nuclear} + \text{Kamuthi Solar} + \text{Muppandal Wind}$$
Clean and Renewable Energy Vanguard

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Gondwana coal is prime bituminous metallurgical coal that constitutes 98 percent of India coal reserves; Tertiary coal is younger, sub-bituminous to lignite, with high moisture and sulfur.

• **Alert:** Digboi in Assam is India oldest operational oil field (drilled in 1889), but Mumbai High is the single largest domestic crude oil producer.

• **Alert:** Monazite beach sands in Kerala are the commercial source of Thorium, whereas Uranium is mined from Jaduguda in Jharkhand.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Distinguish between ferrous metallic minerals and non-ferrous metallic minerals with two examples of each. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Ferrous Minerals:** Metallic minerals that contain iron, forming the metallurgical backbone for steel industries; examples include iron ore, manganese, and chromite. **[1.5 Marks]**

• **Non-Ferrous Minerals:** Metallic minerals that do not contain iron, essential in electrical and engineering industries; examples include bauxite (aluminium ore) and copper. **[1.5 Marks]**

2. Describe the North-Eastern Plateau mineral belt of India, naming its principal mineral deposits and states. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **Geographic Scope:** Covers the Chotanagpur plateau across Jharkhand, Odisha plateau, and parts of West Bengal and Chhattisgarh. **[1 Mark]**

• **Mineral Heartland:** Regarded as the richest mineral belt of India, holding the vast majority of metallurgical coal and iron ore reserves. **[1.5 Marks]**

• **Principal Deposits:** High-grade haematite iron ore, Gondwana coking coal (Jharia, Raniganj, Bokaro), manganese, bauxite, and mica. **[1.5 Marks]**

3. Differentiate between Gondwana coal and Tertiary coal in India regarding age, quality, and geographical distribution. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **Gondwana Coal:** Over 200 million years old; high-grade bituminous metallurgical coal with low moisture and sulfur content; accounts for 98 percent of national reserves; concentrated in Damodar, Mahanadi, and Godavari river valleys. **[2 Marks]**

• **Tertiary Coal:** Approximately 55 million years old; younger, sub-bituminous coal and lignite with high sulfur and moisture content; found in Assam, Meghalaya, Arunachal Pradesh, and Neyveli in Tamil Nadu. **[2 Marks]**

4. Name any four major nuclear power stations of India and their respective home states. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **1. Tarapur:** Located in Maharashtra (India first nuclear power station). **[0.75 Mark]**

• **2. Rawatbhata:** Located near Kota in Rajasthan. **[0.75 Mark]**

• **3. Kalpakkam:** Located in Tamil Nadu. **[0.75 Mark]**

• **4. Narora:** Located in Uttar Pradesh. **[0.75 Mark]**

5. Why is non-conventional energy considered the energy of the future for a developing country like India? [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Renewable and Inexhaustible:** Solar, wind, and biomass energy are inexhaustible and naturally replenishing, unlike exhaustible fossil fuels. **[1 Mark]**

• **Eco-Friendly:** Produces zero greenhouse carbon emissions, combating global climate change and urban air pollution. **[1 Mark]**

• **Decentralized Generation:** Can be established in remote villages and isolated hill tracts without expensive national grid transmission lines. **[1 Mark]**

6. State the location and significance of: (a) Mumbai High, (b) Digboi, and (c) Jaduguda. **[Score: 3 Marks]**

Answer & Key Valuation Steps:

• **(a) Mumbai High:** Located 160 km offshore from Mumbai in the Arabian Sea; discovered in 1973; the single largest domestic crude petroleum producing field. **[1 Mark]**

• **(b) Digboi:** Located in northeastern Assam; India oldest operational oil field, discovered in 1889. **[1 Mark]**

• **(c) Jaduguda:** Located in the Singhbhum thrust belt of Jharkhand; premier commercial uranium mining centre powering India nuclear reactors. **[1 Mark]**

7. Explain any two measures required for the conservation of mineral resources in India. **[Score: 2 Marks]**

Answer & Key Valuation Steps:

• **1. Scrap Recycling and Substitution:** Recycling metal scrap (copper, aluminium, lead) and developing ceramic or polymer substitutes. **[1 Mark]**

• **2. Efficient Mining Technology:** Deploying high-efficiency processing technology to reduce extraction wastage and utilize low-grade ores economically. **[1 Mark]**